The Quant Grimoire

Part I - Measure Theory (based on J.F. Le Gall)

Quentin DAZY

October 31, 2026

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Chapter 1

Measure Spaces

This chapter lays the measure-theoretic foundation required to build the Itô integral. While standard texts like Le Gall focus heavily on pure mathematical rigor, our goal here is dual : to preserve that formal architecture while bridging the gap toward quantitative intuition. Before we can price a derivative or simulate a stochastic process in Python, we must first rigorously define the space of possible events and how we measure them.

> *"The starting idea of measure theory is to assign a non-negative real number (the measure of this subset) to each subset of a given set, in a way that satisfies certain natural additivity properties (the measure of a disjoint union must be the sum of the measures).*
>
> *For deep reasons, it is generally not possible to define the measure of any arbitrary subset, and one must restrict to a certain class of subsets, called* ***measurable sets****; a set equipped with a σ-algebra is called a* ***measurable space****."* — *J.-F. Le Gall [1]*

1.1 Measurable Sets and σ-Algebras

Definition 1.1: σ-Algebra

Let E be a set. A collection $\mathcal{A}$ of subsets of E is called a **σ-algebra** on E if:

(i) Certain Event : $E \in \mathcal{A}$;

(ii) Closed under Complementation : $A \in \mathcal{A} \implies A^c \in \mathcal{A}$;

(iii) Closed under Countable Unions : If $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$, then $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{A}$.

Integration relies on the passage to the limit. We need to sum an infinite number of small pieces to perform limit calculations. The elements of $\mathcal{A}$ are called measurable sets, or $\mathcal{A}$-measurable sets if there is any ambiguity. We say that $(E, \mathcal{A})$ is a **measurable space**.

Grimoire Insight: σ-algebra definition

Grimoire Insight: Intuition: σ-Algebras as Information Resolution

Think of a σ-algebra $\mathcal{F}$ as the **resolution of an image** representing the market:

- **Coarse resolution (Little Information):** If $\mathcal{F} = \{\emptyset, \Omega\}$, the image is completely pixelated into a single block. You only know that "something will happen" (Ω), but you cannot distinguish any specific outcome.
- **Finer resolution (More Information):** As price observations arrive, the image sharpens. The set Ω is partitioned into smaller subsets. An event $A \in \mathcal{F}$ means your current resolution is sharp enough to tell whether the true state of the world lies inside A or not.

In financial modeling, a **filtration** $(\mathcal{F}_t)_{t\geq 0}$ is simply a movie where the picture becomes clearer as time t progresses.

Grimoire Insight: Intuition: σ-Algebras as Decidable Payoffs

Mathematically, an event $A \subseteq \Omega$ belongs to $\mathcal{F}$ if and only if, given our current information, we can answer the question with certainty: *"Did event A occur?"*
In quantitative finance, if a derivative contract pays \$1 when A occurs and \$0 otherwise, the contract can only be settled at time t if $A \in \mathcal{F}_t$. If $A \notin \mathcal{F}_t$, the market does not yet possess enough information to determine the payoff.

1.2 Integration & Limit Theorems

Theorem 1.2: Lebesgue's Dominated Convergence Theorem

Let $(X_n)_{n\geq 0}$ be a sequence of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Assume that:

- $X_n \to X$ almost surely as $n \to \infty$.
- There exists a non-negative random variable $Y \in L^1(\mathbb{P})$ such that $|X_n| \leq Y$ a.s. for all $n \geq 0$.

Then $X \in L^1(\mathbb{P})$ and:

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X]$$

Proof. The proof relies on applying Fatou's Lemma to the non-negative sequence $Y + X_n \geq 0$ and $Y - X_n \geq 0$. □

Example 1.3: Borel σ-Algebra

Let $\Omega = \mathbb{R}$ with the usual Euclidean topology. The Borel σ-algebra $\mathcal{B}(\mathbb{R})$ is the σ-algebra generated by all open sets of $\mathbb{R}$.

Grimoire Insight: Why Borel sets?

We want to measure probabilities of intervals (like a stock price falling between 100 and 110). The Borel σ-algebra is simply the mathematical safety net that ensures we don't accidentally create unmeasurable "monster" sets when taking unions and intersections of these intervals.

Bibliography

[1] Jean-François Le Gall. *Intégration, Probabilités et Processus Aléatoires*. Springer, 2006.